

Nairen Cao

CURRICULUM VITAE

Associate Professor

Institute for Theoretical Computer Science | School of Computing and Artificial Intelligence
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Research Interests

Design and analysis of algorithms for combinatorial optimization; graph and clustering algorithms, including parallel and distributed algorithms; applications to LLM systems, including LLM evaluation and KV-cache optimization.

Academic Appointments

Associate Professor Shanghai University of Finance and Economics Institute for Theoretical Computer Science, School of Computing and Artificial Intelligence	Current
Faculty Fellow Department of Computer Science, NYU Tandon School of Engineering Host: Martin Farach-Colton	Sep. 2024 - Jun. 16, 2026
Postdoctoral Researcher Department of Computer Science, Boston College Host: Hsin-Hao Su	Sep. 2022 - Aug. 2024
Research Assistant Institute for Theoretical Computer Science, Shanghai University of Finance and Economics Host: Zhihao Tang	Mar. 2019 - May 2019
Research Assistant Department of Computer Science, The University of Hong Kong Host: Siu-Ming Yiu	Sep. 2014 - Aug. 2016

Education

Ph.D. in Computer Science, Georgetown University Advisors: Jeremy T. Fineman and Ophir Frieder Dissertation: Single-source Shortest Paths for Digraphs	2016 - 2022
M.S. in Software Engineering, Peking University	2013 - 2016
B.Eng. in Software Engineering, Sun Yat-sen University	2009 - 2013

Honors

Outstanding Paper Award, SPAA 2023 and SPAA 2022.

Conference Publications

- Static to Dynamic Correlation Clustering**
Nairen Cao, Vincent Cohen-Addad, Euiwoong Lee, Shi Li, David Rasmussen, Lolck, Alantha Newman, Mikkel Thorup, Lukas Vogl, Shuyi Yan, Hanwen Zhang
ICALP 2026
- Bounding the Fragmentation of B-Trees Subject to Batched Insertions**
Michael A. Bender, Aaron Bernstein, Nairen Cao, Alexander Conway, Martín Farach-Colton, Hanna Komlós, Yarin Shechter, Nicole Wein
PODS 2026
- Min-Max Correlation Clustering via Neighborhood Similarity**
Nairen Cao, Steven Roche, Hsin-Hao Su
ESA 2025

Conference Publications (continued)

4. **Simultaneously Approximating All Norms for Massively Parallel Correlation Clustering**
Nairen Cao, Shi Li, Jia Ye
ICALP 2025
5. **Solving the Correlation Cluster LP in Sublinear Time**
Nairen Cao, Vincent Cohen-Addad, Euiwoong Lee, Shi Li, David Rasmussen Lolck, Alantha Newman, Mikkel Thorup, Lukas Vogl, Shuyi Yan, Hanwen Zhang
STOC 2025
6. **Understanding the Cluster LP for Correlation Clustering**
Nairen Cao, Vincent Cohen-Addad, Euiwoong Lee, Shi Li, Alantha Newman, Lukas Vogl
STOC 2024
7. **Breaking 3-Factor Approximation for Correlation Clustering in Polylogarithmic Rounds**
Nairen Cao, Shang-En Huang, Hsin-Hao Su
SODA 2024
8. **Parallel and Distributed Exact Single-Source Shortest Paths with Negative Edge Weights**
Vikrant Ashvinkumar, Aaron Bernstein, **Nairen Cao**, Christoph Grunau, Bernhard Haeupler, Yonggang Jiang, Danupon Nanongkai, Hsin-Hao Su
ESA 2024
9. **Nearly Optimal Parallel Algorithms for Longest Increasing Subsequence**
Nairen Cao, Shang-En Huang, Hsin-Hao Su
SPAA 2023, **Outstanding Paper Award**
10. **Parallel Exact Shortest Paths in Almost Linear Work and Square Root Depth**
Nairen Cao, Jeremy T. Fineman
SODA 2023
11. **Nested Active-Time Scheduling**
Nairen Cao, Jeremy T. Fineman, Shi Li, Julián Mestre, Katina Russell, Seeun William Umboh
ISAAC 2022
12. **Parallel Shortest Paths with Negative Edge Weights**
Nairen Cao, Jeremy T. Fineman, Katina Russell
SPAA 2022, **Outstanding Paper Award**
13. **Brief Announcement: Nested Active-Time Scheduling**
Nairen Cao, Jeremy T. Fineman, Shi Li, Julián Mestre, Katina Russell, Seeun William Umboh
SPAA 2022
14. **Brief Announcement: An Improved Distributed Approximate Single Source Shortest Paths Algorithm**
Nairen Cao, Jeremy T. Fineman, Katina Russell
PODC 2021
15. **Efficient Construction of Directed Hopsets and Parallel Approximate Shortest Paths**
Nairen Cao, Jeremy T. Fineman, Katina Russell
STOC 2020
16. **Brief Announcement: Improved Work Span Tradeoff for Single Source Reachability and Approximate Shortest Paths**
Nairen Cao, Jeremy T. Fineman, Katina Russell
SPAA 2020

Conference Publications (continued)

17. **Toward RSA-OAEP without Random Oracles**
Nairen Cao, Adam O'Neill and Mohammad Zaheri
Public Key Cryptography (PKC) 2020
18. **I/O-Efficient Algorithms for Topological Sort and Related Problems**
Nairen Cao, Jeremy T. Fineman, Katina Russell, Eugene Yang
SODA 2019

Journal Publications

19. **Self-supervised Representation Learning on Electronic Health Records with Graph Kernel Infomax**
Hao-Ren Yao, **Nairen Cao**, Katina Russell, Der-Chen Chang, Ophir Frieder, Jeremy T. Fineman
ACM Transactions on Computing for Healthcare (HEALTH) 2024
20. **I/O-Efficient Algorithms for Topological Sort and Related Problems**
Nairen Cao, Jeremy T. Fineman, Katina Russell, Eugene Yang
ACM Transactions on Algorithms 18(1), 2022

Workshop Papers and Posters

21. **Efficient Cost-Aware LLM Evaluation via Bayesian Bandit Gittins Indices**
Qian Xie, Yueli He, **Nairen Cao**
ICML 2026 Workshop on Decision-Making from Offline Datasets to Online Adaptation: Black-Box Optimization to Reinforcement Learning (DEMO), **Spotlight**
22. **Efficient Construction of Directed Hopsets and Parallel Single-source Shortest Paths**
Nairen Cao, Jeremy T. Fineman, Katina Russell
Highlights of Parallel Computing (HOPC), 2023
23. **LRCRYPT: Leakage-Resilient Cryptographic System (Design and Implementation)**
Xiaoqi Yu, **Nairen Cao**, Gongxian Zeng, Ruoqing Zhang, Siu-Ming Yiu
WISA 2016

Teaching

Instructor, NYU Tandon School of Engineering

- CS-GY 6033, Design and Analysis of Algorithms I: Fall 2024, Fall 2025.
- CS-GY 6753, Theory of Computation: Spring 2025, Spring 2026.

Teaching Assistant, Georgetown University

- Database Management Systems: Fall 2019, Spring 2020, Spring 2021 (Ophir Frieder).
- Algorithms: Fall 2020 (Jeremy T. Fineman).

Professional Service

Organizer, NYC Student Theory Day: Spring 2025, Fall 2025, Spring 2026.

Organizer, NYC Graduate Student TCS Day: October 18, 2024.

Reviewer for SPAA, SODA, STOC, FOCS, ITCS, PODC, ICALP, and other conferences.